$$\boxed{\phi 1 \quad S\&R \quad (B2)}$$

$g_{\mu\nu} = g_{\nu\mu}$ $\partial_\mu = \frac{\partial}{\partial x^\mu}$ $\partial_\mu x^\nu = \delta^\nu_\mu$

$g_{\nu\mu} g^{\nu\rho} = \delta^\rho_\mu$ $\partial^\mu = g^{\mu\nu}\partial_\nu$ $\partial_\nu x_\nu = g_{\mu\nu}$

Contravariant $A'^\nu = \Lambda^\nu{}_\mu A^\mu$

Covariant $A'_\nu = (\Lambda^{-1})_\nu{}^\mu A_\mu$

$$\Lambda^\nu{}_\mu = \begin{pmatrix} \gamma & -\gamma\beta & & \\ -\gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

$\frac{dt}{d\tau} = \gamma$ $\left\{ \begin{array}{l} \gamma = \cosh\beta \\ \beta = \tanh\beta \end{array} \right.$

$X^\mu = (ct, \underline{x})$

$U^\mu = \frac{dX^\mu}{d\tau} = \gamma(c, \underline{u})$ $\partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \underline{\nabla}\right)$

$K^\mu = \left(\frac{\omega}{c}, \underline{k}\right)$

$P^\mu = mU^\mu = \left(\frac{E}{c}, \underline{p}\right)$

$\dot{U}^\mu = \frac{dU^\mu}{d\tau} = \gamma^2\left(\frac{\underline{u}\cdot\underline{a}}{c}\gamma^2, \frac{\underline{u}\cdot\underline{a}}{c^2}\gamma^2\underline{u} + \underline{a}\right)$

$F^\mu = \frac{dP^\mu}{d\tau} = \gamma\left(\frac{1}{c}\frac{dE}{dt}, \underline{f}\right)$ $(a_0 = \gamma^3 a)$

$$\left( \frac{dE}{dt} = \underline{f} \cdot \underline{u} \right)$$

$$J^\mu = (\rho c, \underline{j}) \qquad \partial_\mu J^\mu = 0$$

$$\begin{cases} \underline{E} = -\nabla\phi - \dfrac{\partial \underline{A}}{\partial t} \\ \underline{B} = \nabla \times \underline{A} \end{cases} \Rightarrow \begin{cases} \phi = \phi - \dfrac{\partial \chi}{\partial t} \\ \underline{A} = \underline{A} + \nabla\chi \end{cases}$$

(Gauge Transformation)

$$A^\mu = \left(\frac{\phi}{c}, \underline{A}\right) \Rightarrow \underline{A^\mu = A^\mu + \partial^\mu \chi}$$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

$$\begin{cases} \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu \quad (\text{or } \partial_\nu F^{\mu\nu} = \mu_0 J^\mu) \\ \partial^\rho F^{\mu\nu} + \partial^\mu F^{\nu\rho} + \partial^\nu F^{\rho\mu} = 0 \end{cases}$$

Maxwell

$$F'^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma}$$

$$\begin{cases} \underline{E}'_\parallel = \underline{E}_\parallel \\ \underline{B}'_\parallel = \underline{B}_\parallel \\ \underline{E}'_\perp = \gamma(\underline{E}_\perp + \underline{v} \times \underline{B}) \\ \underline{B}'_\perp = \gamma\left(\underline{B}_\perp - \dfrac{1}{c^2}\underline{v} \times \underline{E}\right) \end{cases}$$

from $F^{\mu\nu}$ components.

$$\partial_\mu T^{\mu\nu} = 0$$

$$F^{\mu\nu} = \left(\begin{array}{c|ccc} 0 & & E/c & \\ \hline & 0 & B_z & -B_y \\ -\frac{E}{c} & -B_z & 0 & B_x \\ & B_y & -B_x & 0 \end{array}\right)$$

$$T^{\mu\nu} = \left(\begin{array}{c|c} \varepsilon & \underline{S}/c \\ \hline \underline{S}/c & \sigma_{ij} \end{array}\right)$$

# More on B2 Rev.

by Caroline & Steane

## ① Kinematics

$$\beta = \sqrt{1-\frac{1}{\gamma^2}}$$

$$\frac{d\gamma}{dv} = \gamma^3 \frac{v}{c^2} \qquad \frac{\gamma-1}{\beta^2} = \frac{\gamma^2}{1+\gamma}$$

$$\frac{dt}{d\tau} = \gamma \qquad \frac{d}{dv}(\gamma v) = \gamma^3$$

$$\frac{dt'}{dt} = \gamma_v\left(1-\underline{u}\cdot\underline{v}/c^2\right)$$

Can consider $U\cdot V = U'\cdot V' \longrightarrow \gamma(\omega) = \gamma(u)\gamma(v)\left(1-\frac{\underline{u}\cdot\underline{v}}{c^2}\right)$

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$$\underline{u} = \underline{u}_{\parallel} + \underline{u}_{\perp} \qquad \left(\underline{u}_{\parallel} = (\underline{u}\cdot\underline{v})\,\underline{v}/v^2,\ \ \underline{u}_{\perp} = \underline{u}-\underline{u}_{\parallel}\right)$$

$$\underline{u}'_{\parallel} = \frac{\underline{u}_{\parallel}-\underline{v}}{1-\frac{\underline{u}\cdot\underline{v}}{c^2}} \qquad \underline{u}'_{\perp} = \frac{\underline{u}_{\perp}}{\gamma_v\left(1-\frac{\underline{u}\cdot\underline{v}}{c^2}\right)}$$

(Relativistic transformations of velocities)

$$\tan\theta' = \frac{u'_{\perp}}{u'_{\parallel}} = \frac{u\sin\theta}{\gamma_v(u\cos\theta - v)}$$

where $\tan\theta = \frac{u_{\perp}}{u_{\parallel}} = \frac{u\sin\theta}{u\cos\theta}$

$$\underline{u}' = \underline{u}'_{\perp} + \underline{u}'_{\parallel} = \frac{1}{1-\frac{\underline{u}\cdot\underline{v}}{c^2}}\left(\frac{1}{\gamma_v}\underline{u} - \left(1 - \frac{\underline{u}\cdot\underline{v}}{c^2}\frac{\gamma_v}{1+\gamma_v}\right)\underline{v}\right)$$

$$\tanh\rho = \frac{v}{c} = \beta$$

$\downarrow$

rapidity

$$\cosh\rho = \gamma$$

$$\sinh\rho = \beta\gamma$$

$$\exp(\rho) = \left(\frac{1+\beta}{1-\beta}\right)^{1/2}$$

$$\mathcal{L} = \begin{bmatrix} \gamma & -\gamma\beta & & \\ -\gamma\beta & \gamma & & \\ & & 1 & \\ & & & 1 \end{bmatrix} = \begin{bmatrix} \cosh\rho & -\sinh\rho & & \\ -\sinh\rho & \cosh\rho & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

$$w = \frac{u+v}{1+\frac{uv}{c^2}}$$

$\downarrow$

$$\tanh\rho_w = \frac{\tanh\rho_u + \tanh\rho_v}{1+\tanh\rho_u\tanh\rho_v} = \tanh(\rho_u+\rho_v)$$

$\rightarrow$ $\underline{\rho_w = \rho_u + \rho_v}$

Consider $A' = \mathcal{L}A$

$$A'^T g A' = (\mathcal{L}A)^T g(\mathcal{L}A) = A^T(\mathcal{L}^T g\mathcal{L})A$$

$A^TgA$ is Lorentz invariant as $\underline{\underline{\mathcal{L}^T g\mathcal{L} = g}}$

Four vectors $\quad$ *Convention $(-+++)$

| Symbol | Def. | Components | Invariant |
|---|---|---|---|
| $X$ | $X$ | $(ct, \underline{r})$ | $-c^2\tau^2$ |
| $U$ | $\frac{dX}{d\tau}$ | $(\gamma c, \gamma \underline{u})$ | $-c^2$ |
| $P$ | $m_0 U$ | $(E/c, \underline{p})$ | $-m_0^2 c^2$ |
| $F$ | $\frac{dP}{d\tau}$ | $(\gamma W/c, \gamma \underline{f})$ | |
| $J$ | $\rho_0 U$ | $(c\rho, \underline{j})$ | $-\rho_0^2 c^2$ |
| $A_{(1)}$ | $A$ | $(\phi/c, \underline{A})$ | |
| $A_{(2)}$ | $\frac{dU}{d\tau}$ | $\gamma(\dot{\gamma} c, \dot{\gamma}\underline{u} + \gamma \underline{a})$ | $a_0^2$ |
| $K$ | $K$ | $(\omega/c, \underline{k})$ | |

where $\gamma = \gamma_u \qquad \dot{\gamma} = \frac{d\gamma}{dt} \qquad W = \frac{dE}{dt}$

$$\begin{cases} -\text{ve invariance} \rightarrow \text{timelike} \\ +\text{ve invariance} \rightarrow \text{space-like} \end{cases}$$

Proper time:

$$-c^2dt^2 + dx^2 + dy^2 + dz^2 = -c^2 d\tau^2$$

$$d\tau = dt\left(1 - \frac{u^2}{c^2}\right)^{1/2}$$

$$\rightarrow \frac{dt}{d\tau} = \gamma$$

$$\dot{\gamma} = \frac{d\gamma}{dt} = \gamma^3 \frac{\underline{u}\cdot\underline{a}}{c^2}$$

$$A = \frac{dU}{d\tau} = \gamma \frac{dU}{dt} = \gamma(\dot{\gamma}c, \dot{\gamma}\underline{u} + \gamma\underline{a})$$

$$= \gamma^2\left(\frac{\underline{u}\cdot\underline{a}}{c}\gamma^2, \frac{\underline{u}\cdot\underline{a}}{c^2}\gamma^2\underline{u} + \underline{a}\right)$$

(instantaneous)
In rest frame of particle $\rightarrow$ $A = (0, \underline{a}_0)$

$U \cdot A = 0$ orthogonal 4-vectors

Invariance

$\hookrightarrow$ $\gamma^4\left(-\left(\frac{\underline{u}\cdot\underline{a}}{c}\right)^2\gamma^4 + \left(\frac{\underline{u}\cdot\underline{a}}{c^2}\gamma^2\underline{u} + \underline{a}\right)^2\right) = a_0^2$

$\rightarrow$ $a_0^2 = \gamma^6\left(a^2 - (\underline{u}\times\underline{a})^2/c^2\right)$

If $\underline{u} \perp \underline{a}$ $\rightarrow$ $a_0 = \gamma^2 a$

i.e. circular motion $a_0 = \gamma^2\left(\frac{u^2}{r}\right)$

particle's rest frame

circle's rest frame

If $\underline{u} \parallel \underline{a}$ $\rightarrow$ $a_0 = \gamma^3 a$

- For a null 4-vector (light-like)

$$\cos\theta' = \frac{p_x'}{\left(\frac{E'}{c}\right)} = \frac{\gamma\left(p\cos\theta - \beta\frac{E}{c}\right)}{\gamma\left(\frac{E}{c} - \beta p\cos\theta\right)} = \frac{\cos\theta - \beta}{1 - \beta\cos\theta}$$

$(E = pc)$

$$\left(W = \frac{dE}{dt}\right)$$

## 4-force

$$\underline{F} = \frac{d\underline{P}}{d\tau} = \gamma \frac{d\underline{P}}{dt} = \gamma\left(\frac{d(E/c)}{dt}, \frac{d\underline{p}}{dt}\right)$$

$$= \gamma\left(\frac{W}{c}, \underline{f}\right)$$

## Wave vector

$$\underline{K} = \left(\frac{\omega}{c}, \underline{k}\right)$$

phase $\phi = \underline{K}\cdot\underline{X} = \underline{k}\cdot\underline{r} - \frac{\omega}{c}ct$

$$\phi = \underline{k}\cdot\underline{r} - \omega t$$

Lorentz invariant

CM Frame $\rightarrow$ $\underline{p}_1' + \underline{p}_2' = 0$

$$E^{CM}_{tot} = E'_{tot} = c\sqrt{-(\underline{P}_1' + \underline{P}_2')^2} = c\sqrt{-(\underline{P}_1 + \underline{P}_2)^2}$$

## Doppler Effect

$$\underline{K}\cdot\underline{U} = (\omega/c, \underline{k})\cdot(\gamma c, \gamma\underline{v})$$

$$= \gamma(-\omega + \underline{k}\cdot\underline{v}) = -\gamma\omega\left(1 - \frac{kv}{\omega}\cos\theta\right)$$

$$= -\frac{\omega_0}{c}c = -\omega_0$$

$\hookrightarrow$ source rest frame

$$\omega_0 = \gamma\omega\left(1 - \frac{kV}{\omega}\cos\theta\right)$$ (general) Doppler

## ② Dynamics

4-Force

$$F = \frac{dP}{d\tau} = \gamma\left(\frac{1}{c}\frac{dE}{dt}, \underline{f}\right)$$

$$U \cdot F = -c^2 \frac{dm_0}{d\tau} = 0$$

if rest mass is constant

If rest mass unchanged, the force is called 'pure force'.

$$\rightarrow \frac{dE}{dt} = \underline{f} \cdot \underline{u} \quad (\text{take } U \cdot F)$$

Force transformations:

$$F = \gamma_u\left(\frac{W}{c}, \underline{f}\right)$$

$$\begin{cases} \dfrac{\gamma_{u'}}{c}\dfrac{dE'}{dt'} = \gamma_v\gamma_u\left(\dfrac{1}{c}\dfrac{dE}{dt} - \beta f_{\parallel}\right) \\ \gamma_{u'} f_{\parallel} = \gamma_v\gamma_u\left(-\dfrac{\beta}{c}\dfrac{dE}{dt} + f_{\parallel}\right) \\ \gamma_{u'} f_{\perp} = \gamma_u f_{\perp} \end{cases}$$

Using $\gamma_{u'} = \gamma_v \gamma_u \left(1 - \frac{v \cdot u}{c^2}\right)$

$$f'_{\parallel} = \frac{f_{\parallel} - \left(\frac{v}{c^2}\right)\frac{dE}{dt}}{1 - \underline{u}\cdot\underline{v}/c^2} \qquad f'_{\perp} = \frac{f_{\perp}}{\gamma_v\left(1 - \frac{\underline{u}\cdot\underline{v}}{c^2}\right)}$$

For $u = 0$ (in $S$) $\rightarrow$ $\begin{cases} f'_{\parallel} = f_{\parallel} \\ f'_{\perp} = \frac{f_{\perp}}{\gamma_v} \end{cases}$

In any ref frame.

$$\underline{f} = \frac{d}{dt}(\gamma m \underline{v}) = \gamma m \underline{a} + m \dot{\gamma} \underline{v}$$

for const. $m \rightarrow$ $\begin{cases} f_{\parallel} = \gamma^3 m a_{\parallel} \\ f_{\perp} = \gamma m a_{\perp} \end{cases}$

Uniform B-field $\rightarrow$ $\omega = \frac{qB}{\gamma m}$

**Motion under constant force**

$$\frac{d\underline{p}}{dt} = \underline{f} \rightarrow \underline{p} = \underline{p_0} + \underline{f}t$$

for $\underline{p_0} = 0 \rightarrow$ $p = \gamma m_0 v = ft$

$$V = \frac{ft}{\sqrt{m_0^2 + f^2t^2/c^2}}$$

Let $a_0 = \frac{f}{m_0}$

in IRF of particle

$$V = \frac{dx}{dt} = \frac{a_0 t}{(1 + a_0^2t^2/c^2)^{1/2}} \quad (= c\tanh\theta)$$

$$\left(\frac{a_0 t}{c} = \sinh\theta\right)$$

$$\hookrightarrow \boxed{(x-b)^2 - c^2t^2 = \left(\frac{c^2}{a_0}\right)^2}$$

- Hyperbolic worldline.

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} = \cosh\theta$$

$$\frac{dt}{d\tau} = \gamma = \cosh\theta \qquad \frac{dt}{d\theta} = \frac{c}{a_0}\cosh\theta$$

$$\rightarrow \frac{d\tau}{d\theta} = \frac{c}{a_0} \quad \rightarrow \quad \tau = \frac{c\theta}{a_0}$$

$$a = \frac{dv}{dt} = \frac{dv}{d\theta}\frac{d\theta}{dt} = \frac{c\,\mathrm{sech}^2\theta}{\frac{c}{a_0}\cosh\theta} = \frac{a_0}{\cosh^3\theta}$$

$$a = \frac{a_0}{\gamma^3}$$

## Collisions (Typical)

Use invariance & conservation

& $\underline{v} = \frac{\underline{p} c^2}{E}$

## ③ EM & Tensors

not much different.

Invariants:

$$I_1 = \frac{E^2}{c^2} - B^2$$

$$I_2 = \underline{E} \cdot \underline{B}$$

from $F_{\mu\nu} F^{\mu\nu}$ & $\frac{1}{2} \varepsilon_{\mu\nu\alpha\beta} F^{\alpha\beta} F^{\mu\nu}$

(invariants)

fully antisymmetric tensor